HARP-Select: Confidence-Calibrated Residual Polynomial Experts for Heterophilous Graphs

Anonymous submission

Abstract

Graph neural networks for node classification must often operate on graphs whose neighborhoods mix homophilous smoothing evidence with heterophilous contrastive evidence. This paper studies a simple residual polynomial design for that setting. We first construct HARP-GNN, which combines low-pass propagated features with high-pass residual differences through a node-wise gate. Systematic diagnostics reveal a specialization boundary: self-loop residual filtering is strong on WebKB-style graphs, while an ego-separated no-self variant, HARP-ESep, is much stronger on several larger heterophily benchmarks. We therefore introduce HARP-Select, a validation-calibrated expert selector that trains both residual polynomial experts and selects HARP-ESep only when its validation advantage exceeds a fixed normal-approximation uncertainty threshold. The rule uses validation labels only, is frozen across datasets, and exposes every per-split decision. Across six fixed-split heterophily benchmarks, HARP-Select removes the significant Chameleon and Squirrel deficits of HARP-GNN while preserving its Texas and Wisconsin behavior; robust paired checks indicate branch specialization rather than unbounded leaderboard dominance. On external critical-heterophily graphs, HARP-ESep improves Roman-Empire by 2.75 percentage points under paired tests and the frozen selector routes all splits to it, while conservatively retaining HARP-GNN on Amazon-Ratings. The resulting contribution is an auditable expert-and-routing framework: it records when low/high-pass residual experts help, when they fail, and how validation-calibrated routing can avoid committing to a single brittle graph filter. Anonymous code is available at https://anonymous.4open.science/status/HARP-5D26.

Introduction

The success of message-passing graph neural networks (GNNs) is often explained through a useful low-pass bias: neighboring nodes are averaged so that class-consistent local neighborhoods become easier to classify (Kipf and Welling 2017; Klicpera, Bojchevski, and Günnemann 2019; Wu et al. 2019). This bias is also a liability. In heterophilous or mixed graphs, edges may connect role-complementary, label-dissimilar, or locally ambiguous nodes. Repeated smoothing can then erase discriminative ego features and amplify misleading neighborhoods. Many recent designs respond by concatenating multiple powers of the adjacency matrix, separating ego and neighbor representations, or learning more flexible graph filters (Abu-El-Haija et al. 2019; Zhu et al. 2020; Chien et al. 2021; Bo et al. 2021; He, Wei, and Xu 2021; Lim et al. 2021).

The empirical problem is sharper than a binary homophily-versus-heterophily distinction. Different datasets, and even different fixed splits of the same benchmark family, can prefer different structural priors. In our experiments, a self-loop residual polynomial GNN is competitive on Texas and Wisconsin, but suffers large deficits on Chameleon and Squirrel. An ego-separated no-self residual variant repairs those larger-heterophily deficits, but hurts WebKB. This pattern suggests that a single global filter family is miscalibrated: the useful inductive bias is itself dataset- and split-dependent.

We study this boundary through residual polynomial experts. The first expert, HARP-GNN, builds low-pass bases $\{\hat{A}^k X\}$ and high-pass residual bases $\{\hat{A}^{k-1}X - \hat{A}^k X\}$, then fuses them with a node-wise gate. The second expert, HARP-ESep, follows the lesson of ego-neighbor separation: it encodes ego features first, propagates hidden states with a no-self adjacency, applies the same residual low/high-pass filtering in hidden space, and classifies from concatenated ego and residual-filtered views. Finally, HARP-Select trains both experts and selects the ego-separated expert only when its validation advantage exceeds a fixed confidence threshold.

This formulation deliberately avoids test-time oracle selection. For each split, the selector computes the difference between the two validation accuracies and compares it with 1.96 times a normal-approximation standard error. The threshold is fixed before evaluation, every decision is logged, and test labels do not enter the routing rule. The method is therefore more expensive than a single GNN, but it gives a transparent answer to a common heterophily failure mode: if two structural priors specialize differently, route between them by validation evidence rather than pretending one prior is universally correct.

Our contributions are:

- We formulate residual polynomial low/high-pass graph experts and isolate a concrete self-loop versus ego-separated specialization boundary.

- We introduce HARP-Select, a validation-calibrated selector that routes between the two experts using a frozen uncertainty threshold and no test labels.
- We provide a reproducible fixed-split evaluation with paired tests, bootstrap intervals, exact sign-flip tests, Holm correction, result coverage audits, and manuscript/package verifiers.
- We report a bounded but falsifiable evidence pattern: HARP-Select removes large significant deficits on Chameleon and Squirrel, routes correctly on Roman-Empire, and exposes cases where conservative validation evidence is insufficient to justify switching experts.

Related Work

Low-pass message passing. GCN (Kipf and Welling 2017), GraphSAGE (Hamilton, Ying, and Leskovec 2017), SGC (Wu et al. 2019), APPNP (Klicpera, Bojchevski, and Günnemann 2019), GAT (Velickovic et al. 2018), and GIN (Xu et al. 2019) are representative architectures that propagate information over graph neighborhoods. Their filtering interpretation follows graph signal processing and localized spectral convolution (Hammond, Vandergheynst, and Gribonval 2011; Shuman et al. 2013; Defferrard, Bresson, and Vandergheynst 2016; Ortega et al. 2018). They are strong when connected nodes tend to share labels, but neighborhood aggregation often behaves as a low-pass filter that can over- or mis-smooth under heterophily (Nt and Maehara 2019; Li, Han, and Wu 2018; Oono and Suzuki 2020; Rong et al. 2020; Chen et al. 2020; Yan et al. 2022).

GNNs beyond homophily. MixHop (Abu-El-Haija et al. 2019) concatenates multiple adjacency powers. H2GCN (Zhu et al. 2020) separates ego and higher-order neighborhood information. GPR-GNN (Chien et al. 2021), FAGCN (Bo et al. 2021), and BernNet (He, Wei, and Xu 2021) learn flexible spectral or frequency-aware filters. LINKX (Lim et al. 2021) shows that simple feature and adjacency encoders can be competitive on non-homophilous graphs. ACM, GloGNN, label-wise propagation, and neural sheaf diffusion broaden this space through adaptive channels, global aggregation, label-conditioned messages, and learned transport (Luan et al. 2022; Li et al. 2022; Dai et al. 2022; Bodnar et al. 2022). Spectral support and polynomial-basis analyses further motivate richer frequency responses (Balcilar et al. 2021; Wang and Zhang 2022). Our work is closest in spirit to this family, but focuses on a compact residual high-pass basis and on validation-calibrated selection between two structural priors.

Evaluation under heterophily. Graph benchmarks are sensitive to split protocol, metric choice, and baseline coverage (Shchur et al. 2018; Pei et al. 2020; Lim et al. 2021; Platonov et al. 2023). We therefore treat fixed-split paired comparisons and result audits as part of the method contribution. This emphasis treats statistical auditability as part of the model design rather than as a post-hoc reporting detail.

Model selection and auditability. Many heterophily architectures increase expressiveness inside a single model, for example by learning spectral coefficients, concatenating structural channels, or separating ego and neighborhood encoders. Those choices are valuable, but they can obscure a simpler question: when two plausible structural priors disagree, is there enough validation evidence to prefer one before test labels are observed? HARP-Select makes this model-selection step explicit. The selector is not tuned per dataset, does not search over thresholds, and records the validation margin and uncertainty threshold for every split. This turns branch specialization into an auditable object rather than an after-the-fact explanation of aggregate test scores.

Residual Polynomial Experts

Let $G = (V, E)$ be an undirected graph with feature matrix $X \in \mathbb{R}^{n\times d}$, labels for a subset of nodes, and normalized adjacency operators. For the self-loop expert, let $\hat{A}$ denote the symmetrically normalized adjacency with self-loops. For the ego-separated expert, let $\tilde{A}$ denote the corresponding no-self normalized adjacency. Figure 1 summarizes the two-expert training and validation-calibrated routing pipeline.

Self-Loop Residual Expert

HARP-GNN constructs two signal families. The low-pass bases are

$$L_k = \hat{A}^k X, \quad k = 0, \ldots, K, \tag{1}$$

and the high-pass residual bases are

$$H_k = \hat{A}^{k-1} X - \hat{A}^k X, \quad k = 1, \ldots, K. \tag{2}$$

The branches are mixed with learned polynomial weights:

$$Z_L = \sum_{k=0}^{K} \alpha_k W_L L_k, \quad \alpha = \mathrm{softmax}(\theta_L), \tag{3}$$

$$Z_H = \sum_{k=1}^{K} \beta_k W_H H_k, \quad \beta = \mathrm{softmax}(\theta_H). \tag{4}$$

The residual branch uses differences between consecutive propagation orders. Thus $H_1 = X - \hat{A}X$ directly captures the local contrast that ordinary averaging removes. In the eigenbasis of $\hat{A}$, the low-pass response is a nonnegative polynomial $p_L(\lambda) = \sum_k \alpha_k \lambda^k$, while the residual branch has response $p_H(\lambda) = (1-\lambda)\sum_{k=1}^{K} \beta_k \lambda^{k-1}$. The high-pass response therefore vanishes on the constant component $\lambda = 1$ and concentrates capacity on deviations that smoothing suppresses. This gives a simple diagnostic axis: the model is not only adding parameters, but explicitly comparing a smoothing polynomial with its finite-difference complement.

To fuse the branches locally, HARP-GNN computes a label-free feature-variation statistic

$$r_i = \sigma\left(\frac{\|x_i - \mathrm{mean}_{j\in\mathcal{N}(i)}\, x_j\|_2 - \mu}{s + \epsilon}\right), \tag{5}$$

where μ and s are graph-level mean and standard deviation. A gate $g_i = \mathrm{MLP}_g([x_i, r_i])$ produces feature-wise values in $(0, 1)$, and the hidden representation is

$$Z_i = (1 - g_i) \odot Z_{L,i} + g_i \odot Z_{H,i}, \tag{6}$$

followed by normalization, nonlinearity, dropout, and a linear classifier.

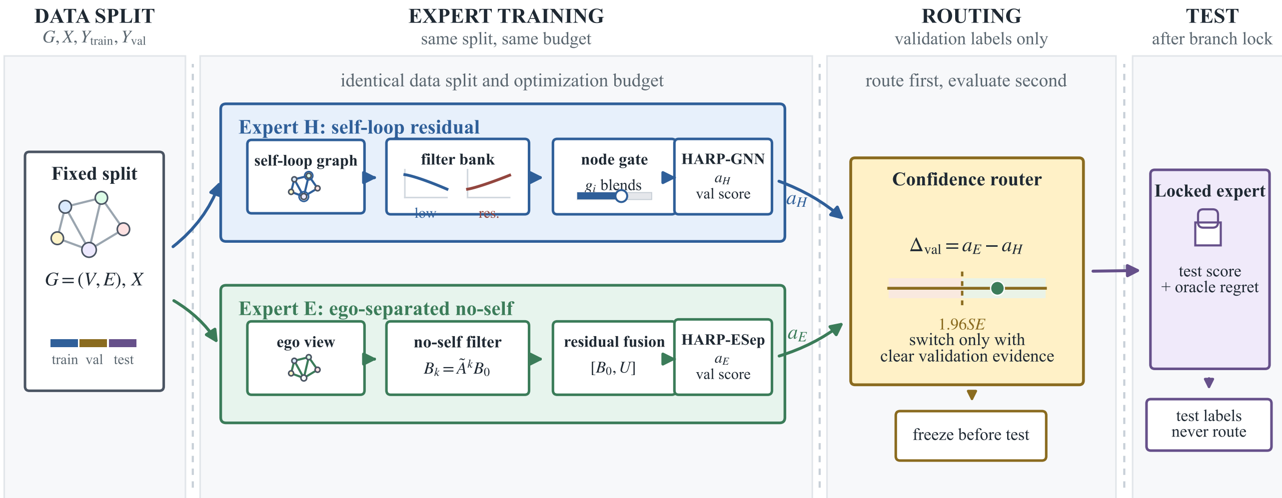

Figure 1: HARP-Select framework. Two residual polynomial experts are trained with the same split and optimization budget. The routing rule uses validation accuracy only and is fixed across datasets; test labels are evaluated only after the branch decision has been made.

Ego-Separated Residual Expert

HARP-ESep uses the same low/high-pass residual idea but changes the structural prior. It first encodes ego features,

$$B_0 = \mathrm{ReLU}(XW_E), \tag{7}$$

then propagates hidden states with the no-self operator:

$$B_k = \tilde{A}^k B_0, \quad k = 1, \dots, K. \tag{8}$$

Low-pass and high-pass hidden views are

$$U_L = \sum_{k=0}^{K} \gamma_k B_k, \quad U_H = \sum_{k=1}^{K} \delta_k (B_{k-1} - B_k), \tag{9}$$

with softmax-normalized weights γ and δ. The gate receives ego, low-pass, high-pass, their absolute difference, and the same optional feature-variation signal:

$$q_i = \mathrm{MLP}_q([B_{0,i}, U_{L,i}, U_{H,i}, |U_{L,i} - U_{H,i}|, r_i]). \tag{10}$$

The residual-filtered view is

$$U_i = (1 - q_i) \odot U_{L,i} + q_i \odot U_{H,i}. \tag{11}$$

HARP-ESep classifies from $[B_{0,i}, U_i]$. This preserves ego evidence explicitly while allowing no-self neighborhood signals to contribute through residual filtering.

HARP-Select: Validation-Calibrated Routing

The two experts specialize in different regimes, so HARP-Select treats branch choice as a validation-calibrated model-selection problem. For each fixed split, both experts are trained independently using the same training labels and optimization budget. Let a_H and a_E be the validation accuracies of HARP-GNN and HARP-ESep, and let n_{val} be the validation-set size. The selector computes

$$\mathrm{SE} = \sqrt{\frac{a_H(1-a_H)}{n_{\text{val}}} + \frac{a_E(1-a_E)}{n_{\text{val}}}}, \tag{12}$$

and chooses HARP-ESep only when

$$a_E - a_H > 1.96\,\mathrm{SE}. \tag{13}$$

Otherwise it retains HARP-GNN. Each reported HARP-Select row logs both branch scores, the validation margin, the threshold, the decision, and oracle regret before any test-label comparison.

The multiplier 1.96 is the conventional two-sided 95% normal critical value and is fixed for all datasets. Because the two validation accuracies are measured on the same validation nodes, the independence approximation can be conservative when errors are positively correlated. This conservatism is intentional: the selector should move to the no-self expert only when validation evidence is clearly larger than sampling noise. The tradeoff is that small but stable improvements may be missed, as observed on Amazon-Ratings.

Why not a learned branch gate? A single end-to-end gate could in principle interpolate between HARP-GNN and HARP-ESep at each node. We tested node-adaptive and logit-blending variants during method development, but they were less reliable: the adaptive branch over-selected the no-self expert on small WebKB graphs and under-selected it on Chameleon and Squirrel, while logit blending was too conservative on the same larger heterophily rows. The validation selector is therefore intentionally coarse. It asks for split-level evidence that one structural prior is better for the dataset and

Table 1: External critical-heterophily dataset statistics for split 0. Split lists train/validation/test counts; edges are undirected after symmetrization; Hom. is edge homophily.

Dataset	$\|V\|$	$\|E\|$	d/C	Split	Hom.
Roman-Emp.	22,662	32,927	300/18	11,331/5,665/5,666	0.047
Amazon-Rat.	24,492	93,050	300/5	12,246/6,123/6,123	0.380

training protocol at hand. This loses node-level flexibility, but it avoids a second learned mechanism whose errors are hard to separate from the graph filter itself. In a review setting, the coarse selector also gives a cleaner falsification target: if validation margins do not justify a branch switch, the method must retain the self-loop expert and accept the resulting oracle regret.

Experiments

Questions. We ask: (Q1) Do residual high-pass bases help on heterophilous graphs? (Q2) Does ego separation repair the larger-heterophily failure mode of the self-loop expert? (Q3) Can a frozen validation-calibrated selector route between the two without using test labels? (Q4) Does the evidence survive paired and robust split-level tests?

Datasets. We evaluate WebKB Texas, Wisconsin, and Cornell; larger Geom-GCN Actor, Chameleon, and Squirrel; and two external critical-heterophily graphs, Roman-Empire and Amazon-Ratings. The first six datasets use the standard 10 fixed Geom-GCN splits (Pei et al. 2020). Roman-Empire and Amazon-Ratings use the fixed masks supplied by the critical-heterophily benchmark suite (Platonov et al. 2023). Table 1 reports the external graph sizes and split statistics used in the external evaluation. For the binary critical-heterophily suite, we implement ROC-AUC support and report complete 10-split Minesweeper and Tolokers branch comparisons; Questions remains a loader and metric sanity check and is excluded from the main empirical claims.

Baselines. The implemented non-HARP baselines are MLP, GCN, SGC, APPNP, MixHop, GPR-GNN, an H2GCN-style separator, and LINKX. For paired comparisons, each dataset is compared against the strongest implemented non-HARP baseline on that dataset. This is not a complete heterophily leaderboard: official-code FAGCN, BernNet, and newer methods are outside the implemented comparison, so broad state-of-the-art claims are intentionally out of scope.

Protocol. All results are averaged over the 10 fixed splits. Hyperparameters are selected by validation accuracy. For HARP-Select, branch selection is performed independently on each split using validation labels only. We report paired t-tests, bootstrap 95% confidence intervals, exact paired sign-flip tests, win/tie/loss counts, and Holm-adjusted p-values where available. Every experiment writes one CSV row per dataset, model, and split, and coverage scripts verify that the expected grid is complete.

Reproducibility controls. The artifact treats the reported tables as generated objects rather than manually edited numbers. Each table in the manuscript is rebuilt from CSV files, and verification scripts check representative means, standard deviations, paired tests, table text, citation keys, anonymous submission markers, and package synchronization. For the selector, the diagnostic CSV stores both branch validation accuracies, the computed threshold in Eq. 13, the selected branch, and the test accuracy evaluated only after the decision is fixed. This is deliberately more bookkeeping than a standard benchmark script, but it reduces two common failure modes in heterophily papers: silently changing split coverage and accidentally converting an oracle branch comparison into a claimed method.

Main Results

Table 2 shows the specialization pattern. On Texas and Wisconsin, the self-loop expert remains strongest among the two HARP branches, so HARP-Select retains HARP-GNN on all splits. On Chameleon and Squirrel, HARP-ESep is much stronger and the selector routes all splits to it. On Actor, the ego-separated branch has a small mean advantage, but its validation gains do not exceed the fixed confidence threshold, so HARP-Select conservatively keeps HARP-GNN. This is the intended behavior: the selector moves only when validation evidence is large relative to the validation-set uncertainty.

The original six-dataset comparison in Table 3 is balanced rather than dominant. HARP-Select is positive against the strongest implemented non-HARP baseline on Texas, Wisconsin, and Squirrel, and negative on Cornell, Actor, and Chameleon. None of the paired t-tests are significant at $p < 0.05$. In particular, current positive margins are not significant at $p < 0.05$. The robust checks in Table 4 lead to the same conclusion: exact sign-flip tests with Holm correction produce no significant row. This matters because the original self-loop expert had significant negative margins on Chameleon and Squirrel; HARP-Select removes those significant deficits, but it does not create a significant superiority claim.

External validation becomes sharper after adding non-HARP baselines. On Roman-Empire, HARP-ESep improves over HARP-GNN by 2.75 percentage points, and Table 5 shows that HARP-Select also exceeds the strongest implemented non-HARP baseline, H2GCN, by 0.79 points with 10/10 split wins and Holm-corrected exact sign-flip $p = 0.008$. The frozen selector routes all Roman-Empire splits to HARP-ESep. On Amazon-Ratings, HARP-ESep has a smaller 0.50 point mean advantage over HARP-GNN that is not significant, while LINKX is the strongest implemented baseline. HARP-Select keeps HARP-GNN on all splits and trails LINKX by 0.91 points, exposing a real limitation: the current expert library lacks a LINKX-like adjacency expert for this graph. The binary critical-heterophily rows in Table 6 sharpen the mechanism rather than giving a uniform gain. On Minesweeper, HARP-ESep reaches 89.64 ROC-AUC versus 88.18 for HARP-GNN, a +1.45 point paired difference with 10/10 split wins. On Tolokers, the direction reverses: HARP-GNN reaches 82.79 while HARP-ESep reaches 79.22, a -3.56 point difference with 0/10 split wins.

Table 2: HARP-Select summary over fixed splits. HARP-Select chooses HARP-ESep only when its validation advantage exceeds the frozen $1.96\,\mathrm{SE}$ threshold. Oracle is reported only as diagnostic regret, not as a method.

Dataset	HARP-GNN	HARP-ESep	HARP-Select	Oracle	ESep splits
actor	35.08 ± 1.36	35.62 ± 1.09	35.08 ± 1.36	35.86 ± 0.92	0/10
amazon-ratings	45.43 ± 0.71	45.94 ± 0.48	45.43 ± 0.71	46.18 ± 0.31	0/10
chameleon	55.90 ± 2.49	63.42 ± 2.24	63.42 ± 2.24	63.42 ± 2.24	10/10
cornell	74.86 ± 4.60	68.92 ± 5.87	74.86 ± 4.60	75.68 ± 4.41	0/10
roman-empire	75.76 ± 0.63	78.51 ± 0.57	78.51 ± 0.57	78.51 ± 0.57	10/10
squirrel	36.66 ± 1.66	46.48 ± 1.10	46.48 ± 1.10	46.48 ± 1.10	10/10
texas	86.49 ± 4.59	75.41 ± 4.84	86.49 ± 4.59	86.49 ± 4.59	0/10
wisconsin	87.45 ± 3.23	82.16 ± 4.57	87.45 ± 3.23	87.45 ± 3.23	0/10

Table 3: HARP-Select versus the strongest implemented non-HARP baseline on the original six fixed-split heterophily datasets. Diff is measured in percentage points over matched splits.

Dataset	Best baseline	Baseline	HARP-Select	Diff (pp)	p
actor	LINKX	35.67	35.08	-0.59	0.245
chameleon	H2GCN	64.21	63.42	-0.79	0.395
cornell	MLP	76.49	74.86	-1.62	0.405
squirrel	LINKX	45.48	46.48	+1.01	0.273
texas	MixHop	84.59	86.49	+1.89	0.132
wisconsin	H2GCN	85.49	87.45	+1.96	0.085

Both rows have Holm-corrected exact sign-flip $p = 0.004$. We treat this as branch evidence only; the validation selector uses an accuracy-derived uncertainty rule and is therefore not applied to ROC-AUC datasets.

Mechanistic Evidence

Figure 2 depicts the internal residual polynomial block used by the self-loop expert.

The ablations in Table 7 support the residual interpretation. Low-pass-only HARP is weak on WebKB, while high-pass residual filtering recovers much of the lost performance. The full feature-wise gate is strongest on Texas and competitive on Wisconsin, although the no-signal variant remains close and is slightly better on Cornell. Thus the safest mechanistic claim is that residual low/high-pass fusion is useful; the handcrafted feature-variation signal is a bounded hypothesis rather than the sole source of the gain. Table 8 further shows that learned filters allocate substantial mass to first-order high-pass residuals on WebKB, consistent with the idea that local contrastive evidence is important under heterophily.

Failure Modes

The strongest negative evidence is as important as the positive branch result. On WebKB, globally replacing HARP-GNN with the no-self expert is harmful, which means ego separation is not a universal cure for heterophily. On Actor, HARP-ESep has a mild mean advantage over HARP-GNN, but the validation margin is too small to trigger the frozen selector; this produces a conservative decision rather than a hidden oracle. On Amazon-Ratings, the same conservatism misses a small non-significant HARP-ESep advantage. Tolokers points in the opposite direction from Minesweeper: under complete 10-split ROC-AUC evaluation, every paired split favors the self-loop expert. These failures clarify the intended use of HARP-Select. It is most appropriate when two structural priors have visibly different validation behavior; it is not a replacement for stronger single-model architectures when validation evidence is weak or when deployment requires one compact model.

Cost

HARP-Select trains both experts before validation routing. The artifact's recorded CPU wall-clock trace gives a macro mean two-expert cost of 95.0 seconds per split across the eight selector datasets, $1.72\times$ the self-loop expert on average and 2.11 hours across all 80 selector branch runs. This is acceptable for an auditable benchmark study but not an efficiency claim. A natural next step is to distill the selected expert into a shared encoder or train a cheaper selector before full expert training.

Limitations and Ethics

This manuscript does not claim final state-of-the-art performance. The main limitation is baseline coverage. The implemented comparison includes MLP, GCN, SGC, APPNP, MixHop, GPR-GNN, an H2GCN-style separator, and LINKX, but not official-code FAGCN, BernNet, or newer 2025–2026 heterophily methods. The external Roman-Empire result is therefore evidence for the HARP branch design, not a complete leaderboard claim. The critical-heterophily ROC-AUC evaluation now includes complete Minesweeper and Tolokers branch comparisons, while Questions remains a loader and metric sanity check rather than a reported main claim.

Table 4: Robust split-level checks for HARP-Select against the strongest implemented non-HARP baseline. W/T/L counts split-level wins, ties, and losses; Holm p adjusts exact sign-flip tests across the six rows.

Dataset	Baseline	Diff (pp)	Bootstrap 95% CI	W/T/L	Sign-flip p	Holm p
actor	LINKX	-0.59	[-1.49, +0.24]	3/1/6	0.242	1.000
chameleon	H2GCN	-0.79	[-2.43, +0.86]	4/0/6	0.383	1.000
cornell	MLP	-1.62	[-5.14, +1.62]	3/3/4	0.359	1.000
squirrel	LINKX	+1.01	[-0.53, +2.72]	6/0/4	0.299	1.000
texas	MixHop	+1.89	[-0.27, +4.05]	5/3/2	0.203	1.000
wisconsin	H2GCN	+1.96	[+0.00, +3.73]	7/1/2	0.125	0.750

Table 5: External critical-heterophily comparison against the strongest implemented non-HARP baseline. Best is the strongest non-HARP mean accuracy on that dataset. Diff is HARP-Select minus that best baseline over paired fixed splits; Holm p is the exact sign-flip value after correction across the two HARP-Select rows.

Dataset	Best non-HARP	Best	HARP-GNN	HARP-ESep	HARP-Select	Diff (pp)	Holm p
amazon-ratings	LINKX	46.34 ± 0.81	45.43 ± 0.71	45.94 ± 0.48	45.43 ± 0.71	-0.91	0.020
roman-empire	H2GCN	77.73 ± 0.68	75.76 ± 0.63	78.51 ± 0.57	78.51 ± 0.57	+0.79	0.008

Table 6: Binary critical-heterophily branch comparisons under ROC-AUC over 10 fixed splits. Diff is measured in ROC-AUC percentage points and equals HARP-ESep minus HARP-GNN.

Dataset	HARP-GNN	HARP-ESep	Diff (pp)	n	p
minesweeper	88.18 ± 0.84	89.64 ± 0.67	+1.45	10	< 0.001
tolokers	82.79 ± 0.83	79.22 ± 0.70	-3.56	10	< 0.001

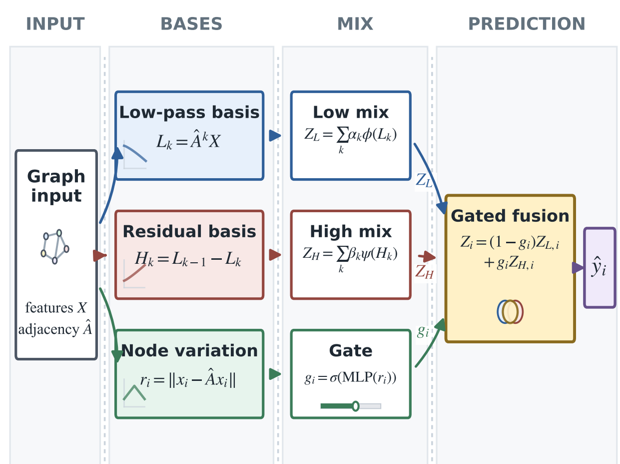

Figure 2: HARP-GNN internal residual polynomial block. HARP-Select uses this self-loop expert as one branch and pairs it with an ego-separated no-self version of the same residual filtering idea.

The selector is conservative by design. This avoids overreacting to validation noise, but misses small advantages such as Amazon-Ratings. It also doubles training cost and selects at the split level rather than learning a single deployable architecture. These are scientific limitations, not incidental engineering details, and we report them explicitly.

This work uses public benchmark datasets for node classification. As with all graph learning systems, deployment on social, educational, or recommendation networks could amplify measurement bias or expose sensitive relational structure. The artifact is intended for benchmark research, and any applied use should include dataset-specific privacy and fairness review.

Practical implications. The experiments suggest a pragmatic workflow for new heterophilous graphs. First, compare a self-loop residual expert and an ego-separated no-self expert on the prescribed validation split instead of choosing a propagation convention by convention. Second, inspect the validation margin together with its uncertainty, because small branch differences can reverse across splits and should not be promoted to architectural claims. Third, report the selector regret against the branch oracle even when the selected model is competitive; regret is the clearest diagnostic of whether the routing threshold is too conservative. Finally, separate branch evidence from selector evidence. Minesweeper and Roman-Empire show that the no-self residual expert can be the better branch, while Tolokers and WebKB warn that the same branch can be worse elsewhere. The scientific value of HARP-Select is precisely this separation: it makes the structural prior, validation decision, and test consequence visible as distinct quantities.

Interpreting mixed evidence. The empirical picture is intentionally uneven. This matters for a heterophily paper because a method that only reports its best branch can obscure the central modeling question: which neighborhood relation is being trusted? HARP-Select makes that question explicit. The WebKB rows show that self information remains important when local labels are noisy but ego features carry

Table 7: WebKB ablations isolating multi-hop mixing, low/high-pass residual branches, and the gate signal.

Dataset	MixHop	HARP-GNN	HARP-Low	HARP-High	HARP-NoSignal
cornell	74.59 ± 3.17	74.86 ± 4.60	51.35 ± 11.18	72.43 ± 6.34	$\mathbf{75.41 \pm 4.12}$
texas	84.59 ± 4.60	$\mathbf{86.49 \pm 4.59}$	62.43 ± 7.69	81.35 ± 6.43	85.14 ± 6.40
wisconsin	83.73 ± 5.47	$\mathbf{87.45 \pm 3.23}$	75.49 ± 6.22	79.02 ± 4.90	$\mathbf{87.45 \pm 4.26}$

Table 8: Learned HARP-GNN polynomial weights on WebKB. Low-pass bases are $L_0, \ldots, L_3$ and high-pass residual bases are $H_1, \ldots, H_3$.

Dataset	L0	L1	L2	L3	H1	H2	H3
cornell	0.317 ± 0.015	0.236 ± 0.017	0.223 ± 0.010	0.223 ± 0.010	0.473 ± 0.054	0.265 ± 0.025	0.263 ± 0.029
texas	0.301 ± 0.017	0.263 ± 0.023	0.218 ± 0.012	0.218 ± 0.010	0.481 ± 0.047	0.262 ± 0.022	0.257 ± 0.025
wisconsin	0.316 ± 0.012	0.227 ± 0.009	0.230 ± 0.008	0.227 ± 0.005	0.459 ± 0.022	0.277 ± 0.016	0.264 ± 0.008

useful class evidence. The Roman-Empire and Minesweeper rows show the opposite risk: retaining the ego state inside every propagation path can dilute contrastive neighborhood evidence. Actor, Chameleon, and Squirrel then act as useful stress tests because neither a self-loop residual expert nor a no-self expert should be expected to repair every benchmark regime. Reading these rows together gives a narrower but more useful claim than a leaderboard statement: branch identity is itself an experimental variable, and freezing the branch decision before test evaluation turns that variable into an auditable part of the method.

Audit trail as a method component. The selector is small enough that each decision can be inspected after the run. For every split we can report the validation margin, its standard-error threshold, the selected branch, the oracle branch, and the resulting regret. This audit trail is not only a reproducibility convenience. It also prevents three common failure modes in graph benchmark papers. First, it separates architecture search from test-set reporting. Second, it exposes datasets where a branch looks promising on average but is too unstable to select confidently on a fixed split. Third, it reveals when the proposed family is simply missing a stronger expert, as in the larger Geom-GCN deficits. Figure 3 makes this audit split-level rather than aggregate. The validation surplus is decisively positive on Squirrel, Chameleon, and Roman-Empire, matching their all-HARP-ESep routing, while Actor and Amazon-Ratings remain close to the decision boundary and are conservatively retained by HARP-GNN. The held-out panel confirms aligned gains in the three switched datasets while exposing borderline cases hidden by mean-only tables. Figure 4 further shows that increasing the confidence multiplier monotonically reduces HARP-ESep selections; the frozen $z = 1.96$ rule lies in a low-regret plateau rather than at a test-tuned optimum.

Conclusion

HARP-Select treats heterophilous graph learning as a validation-calibrated choice between residual polynomial experts rather than a commitment to one universal filter. The self-loop expert is useful on WebKB; the ego-separated expert repairs Chameleon and Squirrel failures and improves Roman-Empire; the frozen selector routes without test labels. Split-level audits distinguish decisive switches from borderline cases, while the threshold sweep shows that $z = 1.96$ is a conservative low-regret operating point rather than a test-tuned optimum. The evidence is deliberately bounded and supports this auditable specialist-and-routing formulation.

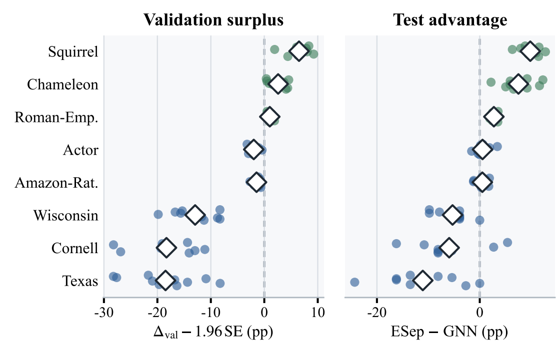

Figure 3: Split-level selector audit over eight accuracy datasets. Positive validation surplus triggers HARP-ESep; test advantage is inspected only after routing. Diamonds denote means.

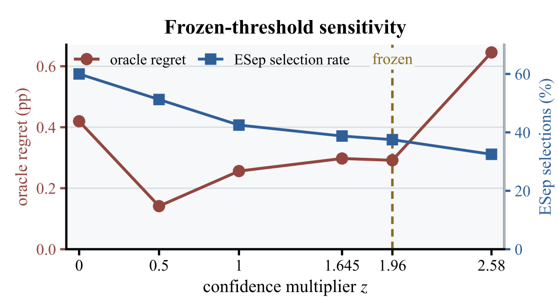

Figure 4: Post-hoc sensitivity over 80 accuracy splits. The submitted $z = 1.96$ point is frozen; the sweep is diagnostic only.

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